

HDOC Day-26

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Batch - 15

1. Algorithm

Main Program

1. Start.
2. Declare `choice`.
3. Display the menu:
 - Right-aligned increasing triangle
 - Right-aligned increasing number triangle
 - Right-aligned decreasing triangle
 - Exit
4. Read `choice`.
5. Use `switch(choice)`.
6. Call the corresponding parameterless function.
7. If the choice is invalid, display an error message.
8. Repeat until the user selects Exit.
9. Stop.

Algorithm: Right-Aligned Increasing Triangle

1. Start function.
2. Input number of rows `n`.
3. If `n <= 0`, display an error message and return.
4. Repeat `i` from 1 to `n`.
5. Print the required leading spaces using a loop from 1 to `n-i`.
6. Print `i` asterisks.
7. Move to the next line.
8. Return.

This corresponds to Activity 1 in the assignment.

Algorithm: Right-Aligned Increasing Number Triangle

1. Start function.
2. Input number of rows `n`.
3. If `n <= 0`, display an error message and return.
4. Repeat `i` from 1 to `n`.
5. Print leading spaces for right alignment.
6. Repeat `j` from 1 to `i`.
7. Print `j`.

8. Move to the next line.
9. Return.

The assignment requires each row to contain natural numbers from 1 up to its row number.

Algorithm: Right-Aligned Decreasing Triangle

1. Start function.
2. Input number of rows n .
3. If $n \leq 0$, display an error message and return.
4. Repeat i from n down to 1.
5. Print $n-i$ leading spaces.
6. Print i asterisks.
7. Move to the next line.
8. Return.

This produces the decreasing pattern specified in Activity 3.

2. Test-Case Table

Test Case	Choice	Input	Expected Output	Actual Output	Result
1	1	$n = 4$	4-row increasing * triangle	Same as expected	Pass
2	2	$n = 4$	Rows 1, 1 2, 1 2 3, 1 2 3 4	Same as expected	Pass
3	3	$n = 4$	4-row decreasing * triangle	Same as expected	Pass
4	1	$n = 1$	*	Same as expected	Pass
5	2	$n = 0$	Invalid number of rows	Same as expected	Pass
6	3	$n = -2$	Invalid number of rows	Same as expected	Pass
7	5	—	Invalid menu choice	Same as expected	Pass
8	4	—	Program terminates	Program terminates	Pass

3. C Program

```
#include <stdio.h>

void increasingTriangle(void);
void numberTriangle(void);
void decreasingTriangle(void);

int main(void)
{
```

```

int choice;

do
{
    printf("\n===== HDOC Day-26 =====\n");
    printf("1. Right-aligned increasing triangle\n");
    printf("2. Right-aligned increasing number
triangle\n");
    printf("3. Right-aligned decreasing triangle\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            increasingTriangle();
            break;

        case 2:
            numberTriangle();
            break;

        case 3:
            decreasingTriangle();
            break;

        case 4:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid menu choice!\n");
    }

} while (choice != 4);

return 0;
}

/* 1. Right-aligned increasing triangle */
void increasingTriangle(void)
{
    int n, i, j;

```

```

printf("Enter number of rows: ");
scanf("%d", &n);

if (n <= 0)
{
    printf("Invalid number of rows!\n");
    return;
}

for (i = 1; i <= n; i++)
{
    /* Print spaces */
    for (j = 1; j <= n - i; j++)
    {
        printf("  ");
    }

    /* Print stars */
    for (j = 1; j <= i; j++)
    {
        printf("* ");
    }

    printf("\n");
}
}

/* 2. Right-aligned increasing number triangle */
void numberTriangle(void)
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid number of rows!\n");
        return;
    }

    for (i = 1; i <= n; i++)
    {

```

```

        /* Print spaces */
        for (j = 1; j <= n - i; j++)
        {
            printf("  ");
        }

        /* Print numbers */
        for (j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }

        printf("\n");
    }
}

/* 3. Right-aligned decreasing triangle */
void decreasingTriangle(void)
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid number of rows!\n");
        return;
    }

    for (i = n; i >= 1; i--)
    {
        /* Print spaces */
        for (j = 1; j <= n - i; j++)
        {
            printf("  ");
        }

        /* Print stars */
        for (j = 1; j <= i; j++)
        {
            printf("* ");
        }
    }
}

```

```

        printf("\n");
    }
}

```

4. Execution

Choice 1

```

[aaryanrajput@aaryans-macbook ~ % nano day26.c
[aaryanrajput@aaryans-macbook ~ % clang day26.c -o day26.c
[aaryanrajput@aaryans-macbook ~ % ./day26.c

===== HDOC Day-26 =====
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 1
Enter number of rows: 4
      *
     * *
    * * *
   * * * *

```

Choice 2

```

===== HDOC Day-26 =====
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 2
Enter number of rows: 4
      1
     1 2
    1 2 3
   1 2 3 4

```

Choice 3

```

===== HDOC Day-26 =====
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 3
Enter number of rows: 4
* * * *
 * * *
  * *
   *

```